

Cellivo Software Requirements Specification

Cloud POS and shop management platform for mobile phone retailers and repair centres, including the shop admin dashboard and the platform super-admin dashboard.

Product	Cellivo (cellivo.com)
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How this document was prepared. Customer-facing features were taken from the public pages of cellivo.com (home, features and pricing). The internal behaviour of each module and the whole of the platform admin dashboard (section 3.16) are specified from standard SaaS POS practice, because they are not visible from the public site. Each requirement is marked *Site* (stated publicly) or *Derived* (proposed). The product team should confirm derived items before sign-off.

1 Introduction

1.1 Purpose of this document

This SRS defines the functional and non-functional requirements for Cellivo, a multi-tenant, cloud-based point-of-sale and business management system built for mobile phone shops, accessory sellers and repair centres. It is the shared reference for developers, testers, designers, project managers, support staff and the business owner, and can serve as the baseline agreement for what the system must do.

1.2 Scope of this document

Cellivo consists of four connected subsystems, all covered by this document:

Subsystem	Address	Main users	Purpose
Public website	cellivo.com	Visitors, prospects	Marketing, feature pages, pricing, blog, contact, demo booking, affiliate sign-up
Account & billing portal	account.cellivo.com	Shop owners	Sign-up, trial, plan purchase, invoices, plan changes, cancellation
Shop application (tenant)	Cloud web app	Owners, managers, cashiers, technicians, accountants	POS billing, IMEI inventory, repairs, customers, finance, staff, branches, reports, including the shop admin dashboard
Platform admin dashboard	Internal (e.g. admin.cellivo.com)	Cellivo super admins, support, sales, content editors	Manage all tenant shops, subscriptions, payments, plans, content, affiliates, support and system health

Out of scope: native desktop installers, hardware manufacture, and the internals of third-party services (SMS gateways, payment processors, WooCommerce), except for how Cellivo connects to them.

1.3 Overview

Cellivo replaces spreadsheets, handwritten bills and generic POS tools in phone shops. It tracks each handset by IMEI or serial number from purchase to sale, return or warranty claim; runs repair jobs from intake to pickup; bills sales, repairs, trade-ins and wholesale orders; and gives owners live sales, stock and profit figures across one or many branches. The product is sold as a subscription in

five tiers (Lite, Starter, Growth, Business, Unlimited) with monthly, yearly and lifetime billing and a free trial that needs no credit card.

2 General description

2.1 Product perspective

Cellivo is a standalone multi-tenant SaaS product. Each subscribing business is a tenant with its own isolated data, one or more branches and its own users. A shared platform layer handles sign-up, billing, plan limits and administration.

From	To	Relationship
Visitor	Public website	Browses, compares plans, starts trial
Public website	Account & billing portal	Sign-up and plan purchase
Account & billing portal	Shop application	Provisions the tenant
Shop application	Branches 1..n	Each branch has its own staff and stock
Shop application	SMS, email, WooCommerce, printers, scanners	Two-way integrations
Platform admin dashboard	Portal and shop application	Manages tenants, plans and payments
End customer	Shop application	Checks repair status by link

2.2 User classes and characteristics

User class	Description	Typical skill
Visitor / prospect	Browses the website, compares plans, books a demo or starts a trial	General web user
Shop owner (tenant admin)	Owens the subscription; full access to all branches, settings, staff and finance	Business owner, low to medium IT skill
Branch manager	Runs one or more branches; approves discounts, transfers and returns	Medium
Cashier / sales staff	Creates sales invoices, takes payments, looks up customers	Low; needs a fast, simple screen
Technician	Works repair jobs, updates status, records parts used	Technical, low software skill

User class	Description	Typical skill
Accountant / finance user	Manages expenses, cheques, bank accounts, ledgers, payroll	Medium
End customer	Receives invoices and SMS, checks repair status by link	General public
Affiliate	Refers new shops and earns commission	General web user
Platform super admin	Cellivo staff with full control over all tenants and platform settings	High
Platform support / sales / content staff	Limited platform roles for tickets, leads, blog and website content	Medium

2.3 Subscription plans and feature gating

Plan limits must be enforced by the system, not only shown on the pricing page.

Capability	Lite	Starter	Growth	Business	Unlimited
List price (USD / month)	10	25	40	55	Custom
Branches	1	1	3	6	Unlimited
Billing, inventory, accessories, customers, suppliers, purchasing	Yes	Yes	Yes	Yes	Yes
IMEI-based stock control	No	Yes	Yes	Yes	Yes
Repair job management	No	Yes	Yes	Yes	Yes
Credit sales, returns, wholesale	No	Yes	Yes	Yes	Yes
Staff roles, advanced reports, profit analytics, full dashboard	Basic reports only	Yes	Yes	Yes	Yes
Priority support	No	Yes	Yes	Yes	Yes

Billing terms: monthly, yearly (two months free) and lifetime. Prices shown in USD by default, with LKR available.

2.4 Operating environment

Cellivo runs in current versions of Chrome, Edge, Safari and Firefox on Windows, macOS, Android, iOS and iPadOS, on desktop, tablet and phone screens. It needs an internet connection and no local installation. Peripherals include USB or Bluetooth barcode scanners, thermal receipt printers (58 mm and 80 mm), A4/A5 printers and cash drawers triggered through the receipt printer.

2.5 Assumptions and dependencies

The system depends on a third-party SMS gateway, a transactional email service, an online payment gateway for subscriptions, cloud hosting with managed backups, and the WooCommerce REST API for store sync. It assumes shops have a stable internet connection during trading hours; offline behaviour is addressed in section 6.

3 Functional requirements

Requirements are ranked by priority.

Must = required for release; **Should** = important; **Could** = nice to have.

3.1 Public website

ID	Requirement	Priority	Source
WEB-01	Show a home page with product summary, feature highlights, customer logos, testimonials, FAQ and calls to start a free trial or book a demo.	Must	Site
WEB-02	Provide a features overview page and dedicated pages for POS billing, IMEI tracking, inventory, repairs, multi-branch, cash drawer, customer loyalty, banking and cheques, staff and payroll, integrations and security.	Must	Site
WEB-03	Provide a pricing page with monthly, yearly and lifetime toggles, USD/LKR currency switch defaulting by visitor location, plan cards and a feature comparison table.	Must	Site
WEB-04	Each plan's "Start free" button must pass the plan code and billing term to the account portal.	Must	Site
WEB-05	Provide a contact page with a form (name, shop name, phone, email, message), phone, email and address; submissions create a lead in the admin dashboard and send an acknowledgement email.	Must	Site / Derived
WEB-06	Provide a "Book live demo" form with preferred date and time; bookings appear in the admin lead list.	Should	Site / Derived
WEB-07	Provide a blog with listing, categories, search and article pages managed from the admin CMS.	Should	Site
WEB-08	Provide About, Affiliate Program, Privacy Policy, Terms of Service and Refund Policy pages, and a link to the support centre.	Must	Site
WEB-09	Provide industry and regional landing pages (e.g. POS for small mobile shops, POS system Sri Lanka).	Could	Site
WEB-10	Every page must have editable title, meta description, canonical URL, Open Graph and Twitter card tags, and be listed in an XML sitemap.	Must	Site
WEB-11	Show a click-to-call phone number and optional WhatsApp chat button on all pages.	Should	Site / Derived

3.2 Sign-up, trial and subscription

ID	Requirement	Priority	Source
ACC-01	Let a new owner register with name, shop name, email, phone, country and password, without a credit card.	Must	Site
ACC-02	Verify the email address (and optionally the phone number by OTP) before the trial starts.	Must	Derived
ACC-03	Provision a new tenant automatically within 60 seconds of verification: default branch, owner account, default roles, tax settings, currency and sample categories.	Must	Site
ACC-04	Run a free trial of configurable length (set by platform admin) on the selected plan, with reminder emails 3 days and 1 day before expiry.	Must	Site / Derived
ACC-05	Let the owner buy or change plan and billing term (monthly, yearly, lifetime) and pay online by card or local methods; issue a tax invoice for each payment.	Must	Site
ACC-06	Support upgrades immediately with proration and downgrades at the next renewal; block a downgrade that would exceed the new plan's branch limit until branches are archived.	Must	Derived
ACC-07	Let the owner cancel at any time; the account stays active until the paid period ends, then becomes read-only for a grace period (e.g. 30 days) before archival.	Must	Site / Derived
ACC-08	Retry failed renewals automatically, notify the owner, and suspend the shop after a configurable number of days unpaid.	Must	Derived
ACC-09	Show billing history, download invoices, update payment method and apply coupon or affiliate codes.	Should	Derived
ACC-10	Let the owner export all shop data (CSV/Excel) before cancelling.	Should	Derived

3.3 Login, access control and security

ID	Requirement	Priority	Source
SEC-01	Users log in with email or username and password; passwords are hashed with a modern algorithm (bcrypt or Argon2).	Must	Site
SEC-02	Support password reset by email link that expires after 30 minutes.	Must	Derived
SEC-03	Support two-factor authentication (authenticator app or SMS OTP), which the owner can make mandatory for chosen roles.	Must	Site

ID	Requirement	Priority	Source
SEC-04	Provide role-based access control with default roles (Owner, Branch Manager, Cashier, Technician, Accountant) and custom roles with per-module view, create, edit, delete and approve permissions.	Must	Site
SEC-05	Hide cost price and profit margin from roles without the "view margins" permission.	Must	Site
SEC-06	Allow the owner to restrict logins to listed IP addresses or ranges.	Should	Site
SEC-07	Allow the owner to approve devices; a login from an unapproved device requires owner approval.	Should	Site
SEC-08	Log out idle sessions after a configurable time; show active sessions and allow remote logout.	Must	Site
SEC-09	Record an audit log entry for every create, edit, delete, price override, discount approval, login and permission change, with user, time, IP, device, and before/after values. Audit logs cannot be edited.	Must	Site
SEC-10	Lock an account for 15 minutes after 5 failed login attempts.	Must	Derived

3.4 Shop admin dashboard

The first screen after login for owners and managers. Cashiers and technicians see a reduced dashboard suited to their role.

ID	Requirement	Priority	Source
DSH-01	Show today's revenue, net profit (after commissions), profit margin and order count, each with change versus the previous period.	Must	Site
DSH-02	Show phones in stock, items below reorder level, and out-of-stock items.	Must	Site
DSH-03	Show repair queue counts by status, including jobs ready for pickup and overdue jobs.	Must	Site
DSH-04	Show a revenue vs. profit chart for the selected period (day, week, month, custom).	Must	Site
DSH-05	Show latest sales, reorder alerts, overdue customer balances and cheques maturing this week.	Should	Site / Derived
DSH-06	Show staff and branch performance leaderboards, best-selling and slow-moving products.	Should	Site

ID	Requirement	Priority	Source
DSH-07	Provide a branch selector (single branch or all branches) for multi-branch tenants.	Must	Site
DSH-08	Provide quick actions: New sale, New repair, Add stock, Add customer.	Must	Site
DSH-09	Show in-app notifications and platform announcements (e.g. planned maintenance, trial expiry).	Should	Derived

3.5 POS billing

ID	Requirement	Priority	Source
POS-01	Add items to a sale by barcode scan, IMEI/serial scan, name search or category browse.	Must	Site
POS-02	When a phone is added, require selection of a specific in-stock IMEI/serial; the same IMEI cannot be sold twice.	Must	Site
POS-03	Calculate line totals, discounts, tax and grand total automatically.	Must	Site
POS-04	Apply line or invoice discounts by amount or percentage within the user's role limit; above the limit, require manager approval by PIN or remote approval.	Must	Site
POS-05	Accept cash, card, bank transfer, cheque, loyalty points, customer credit and split payments; calculate change due.	Must	Site
POS-06	Look up or quick-create a customer at checkout and show history, credit balance and loyalty points.	Must	Site
POS-07	Handle trade-ins: record the incoming device (model, IMEI, condition, value), deduct its value from the sale, and add it to stock as used.	Must	Site
POS-08	Support wholesale price tiers and bulk invoicing for B2B customers (Starter and above).	Should	Site
POS-09	Support credit sales and installment plans with due dates (Starter and above).	Must	Site
POS-10	Process returns and exchanges against the original invoice, restock the IMEI, and refund by original or chosen method (Starter and above).	Must	Site
POS-11	Print thermal receipts or A4/A5 invoices with shop logo, warranty terms and IMEI; email or SMS a digital copy automatically.	Must	Site
POS-12	Hold, resume and void draft sales; voiding a completed invoice requires permission and a reason.	Must	Derived

ID	Requirement	Priority	Source
POS-13	Generate quotations that can be converted into invoices.	Could	Derived
POS-14	Record the salesperson on each invoice for commission calculation.	Must	Site
POS-15	Sell services such as mobile top-ups and bill payments as non-stock items.	Could	Site

3.6 Inventory and IMEI / serial tracking

ID	Requirement	Priority	Source
INV-01	Maintain products with brand, model, category, variant (storage, colour, RAM), condition (new, used, refurbished), cost, selling price, wholesale price, tax class and reorder level.	Must	Site
INV-02	Record each phone by IMEI 1, IMEI 2 and/or serial number; validate IMEI length and Luhn check digit and reject duplicates.	Must	Site
INV-03	Show the full life history of an IMEI: purchase, supplier, transfers, sale, customer, returns, repairs and warranty claims.	Must	Site
INV-04	Track serialized accessories and spare parts in the same way as phones.	Should	Site
INV-05	Track non-serialized accessories (cases, chargers, cables, protectors) by quantity.	Must	Site
INV-06	Generate, print and scan barcodes and labels in configurable sizes.	Must	Site
INV-07	Raise low-stock alerts in-app and by email when quantity falls below the reorder level.	Must	Site
INV-08	Record stock adjustments (damage, loss, found, internal use) with reason and approval.	Must	Derived
INV-09	Run stock counts by scanning, compare with system stock, and post the differences.	Should	Derived
INV-10	Import products and opening stock from Excel/CSV with a validation preview.	Must	Derived
INV-11	Value stock using weighted average cost for quantity items and actual cost for IMEI items.	Must	Derived

3.7 Purchasing and suppliers

ID	Requirement	Priority	Source
PUR-01	Maintain supplier profiles with contact details, payment terms, purchase history and performance notes.	Must	Site
PUR-02	Create, email and track purchase orders.	Must	Site
PUR-03	Receive goods by GRN against a purchase order, capturing IMEIs/serials by scan, and flag quantity or price differences.	Must	Site
PUR-04	Record supplier bills, partial and full payments, and keep a supplier ledger with running balance.	Must	Site
PUR-05	Record purchase returns to suppliers, removing the IMEIs from stock.	Should	Derived
PUR-06	Schedule supplier payments and remind the user before due dates.	Should	Site

3.8 Repair management

ID	Requirement	Priority	Source
REP-01	Create a repair ticket with customer, device model, IMEI, reported fault, accessories received, condition notes, photos, passcode (stored encrypted), estimated cost, advance paid and promised date.	Must	Site
REP-02	Print a job card and customer receipt with ticket number and QR code.	Must	Derived
REP-03	Assign or reassign a technician, showing each technician's current workload.	Must	Site
REP-04	Move tickets through configurable stages (default: Received, Diagnosing, Awaiting approval, Awaiting parts, In progress, Ready for pickup, Delivered, Cancelled) with a timestamped history.	Must	Site
REP-05	Record spare parts used; deduct them from inventory automatically and include their cost in job profit.	Must	Site
REP-06	Generate a repair invoice with parts and labour breakdown, deducting any advance.	Must	Site
REP-07	Send SMS/email to the customer when status changes, based on per-stage templates.	Must	Site
REP-08	Provide a public repair status page reached by a unique link or ticket number plus phone number, showing status only, no internal notes.	Should	Site
REP-09	Set a warranty period on each completed repair and alert before expiry; link warranty returns to the original job.	Should	Site
REP-10	Flag overdue tickets past the promised date on the dashboard.	Must	Site

ID	Requirement	Priority	Source
REP-11	Record customer approval of the quotation before work begins.	Should	Derived

3.9 Customers, credit and loyalty

ID	Requirement	Priority	Source
CRM-01	Maintain customer profiles with name, phone, email, address, NIC/ID (optional), type (retail or wholesale) and notes.	Must	Site
CRM-02	Show each customer's sales, repairs, devices owned (by IMEI), payments and outstanding balance.	Must	Site
CRM-03	Keep a credit ledger with credit limit, due dates and aging (current, 30, 60, 90+ days).	Must	Site
CRM-04	Create installment plans with number of installments, amounts and due dates, and record installment payments.	Should	Site
CRM-05	Send automatic SMS payment reminders before and after due dates.	Should	Site
CRM-06	Run a loyalty programme: configurable points per amount spent, redemption value, expiry, and redemption at checkout.	Should	Site
CRM-07	Import customers from CSV and merge duplicates by phone number.	Could	Derived

3.10 Finance and banking

ID	Requirement	Priority	Source
FIN-01	Open and close cash drawer sessions per user and terminal with opening float, cash in/out entries, expected vs. counted cash, and variance report.	Must	Site
FIN-02	Maintain multiple bank accounts with balances and transactions, including deposits and transfers from the cash drawer.	Must	Site
FIN-03	Record issued and received cheques with number, bank, maturity date and status (pending, deposited, cleared, bounced); alert on maturity.	Must	Site
FIN-04	Maintain customer and supplier ledgers that update automatically from sales, purchases and payments.	Must	Site
FIN-05	Record expenses by category with attachments, and other income (services, top-ups).	Must	Site
FIN-06	Reconcile bank statements (manual matching or CSV import) against system transactions.	Should	Site

ID	Requirement	Priority	Source
FIN-07	Configure tax rates (e.g. VAT/SSCL) and include tax summaries on invoices and reports.	Must	Derived

3.11 Staff, commissions and payroll

ID	Requirement	Priority	Source
HR-01	Add, edit and deactivate staff with role, branch access, contact details and salary information.	Must	Site
HR-02	Define commission rules by staff, product, category, repair or service (fixed or percentage) and calculate commissions automatically.	Should	Site
HR-03	Schedule shifts and record attendance (clock in/out) per branch.	Should	Site
HR-04	Run monthly payroll with basic salary, commissions, allowances, deductions and salary advances, and generate payslips.	Should	Site
HR-05	Record salary advances and deduct them automatically in the next payroll.	Should	Site

3.12 Multi-branch management

ID	Requirement	Priority	Source
BR-01	Create and manage branches with their own name, address, logo, invoice prefix, tax settings, staff and stock, up to the plan limit.	Must	Site
BR-02	Transfer stock between branches with request, approval, dispatch and receipt steps, tracking each IMEI in transit.	Must	Site
BR-03	Let authorised users view real-time stock at any branch.	Must	Site
BR-04	Provide a central dashboard and branch-wise comparison reports.	Must	Site
BR-05	Assign staff roles and permissions per branch.	Must	Site

3.13 Reports and insights

ID	Requirement	Priority	Source
RPT-01	Sales reports by day, week, month or custom range, grouped by product, category, brand, staff, branch and payment method.	Must	Site
RPT-02	Profit reports per phone, accessory, repair job, salesperson and branch.	Must	Site
RPT-03	Stock reports: valuation, aging, dead stock, movement history, IMEI register.	Must	Site

ID	Requirement	Priority	Source
RPT-04	Repair reports: turnaround time, technician productivity, repair revenue and profit.	Must	Site
RPT-05	Customer reports: top customers, purchase frequency, credit exposure, loyalty balances.	Should	Site
RPT-06	Finance reports: profit and loss, cash flow, expense trends, day-end summary, tax summary, bank reconciliation.	Must	Site
RPT-07	KPI dashboard with charts, filterable by branch and date.	Should	Site
RPT-08	Export any report to PDF and Excel; print any report.	Must	Site
RPT-09	Email scheduled reports (e.g. daily summary to the owner).	Could	Derived
RPT-10	Restrict Lite plan to basic reports (sales, stock, customer and supplier lists).	Must	Site

3.14 Integrations and notifications

ID	Requirement	Priority	Source
INT-01	Sync products, stock levels and orders two-way with a WooCommerce store using the owner's API keys.	Should	Site
INT-02	Send SMS for invoices, repair updates and payment reminders through a configured SMS gateway, deducting from the shop's SMS credit balance.	Must	Site
INT-03	Send email invoices, reports and notifications from the platform using the shop's name and reply-to address.	Must	Site
INT-04	Print to thermal (58/80 mm) and A4/A5 printers through the browser print dialog, with an optional silent-print helper.	Must	Site
INT-05	Accept input from USB and Bluetooth barcode scanners working in keyboard mode.	Must	Site
INT-06	Allow owners to edit SMS and email templates with merge fields (customer name, ticket number, amount, link).	Should	Derived
INT-07	Provide an optional WhatsApp message link for invoices and repair updates.	Could	Derived

3.15 Shop settings

ID	Requirement	Priority	Source
SET-01	Business profile: name, logo, address, registration and tax numbers, currency, time zone, date format, language.	Must	Derived
SET-02	Invoice and receipt templates: header, footer, warranty terms, numbering prefixes per branch.	Must	Derived
SET-03	Master data: categories, brands, models, variants, units, repair fault types, repair stages, expense categories, payment methods.	Must	Site
SET-04	Discount limits per role and approval rules.	Must	Site
SET-05	Backup and restore: view daily backups and request a restore to a chosen date (owner only).	Must	Site
SET-06	Subscription panel showing plan, renewal date, branch usage, SMS credits, and upgrade link.	Must	Derived

3.16 Platform admin dashboard (super admin)

The internal back office used by Cellivo staff to run the business. None of it is visible to shop users. All items in this section are derived.

ID	Requirement	Priority
Overview		
ADM-01	Show platform KPIs: total tenants, active paid, on trial, suspended, cancelled; new sign-ups today/week/month; trial-to-paid conversion rate.	Must
ADM-02	Show revenue KPIs: MRR, ARR, lifetime-plan revenue, revenue by plan and by country/currency, churn rate, ARPU, failed payments.	Must
ADM-03	Show charts of sign-ups, conversions, revenue and churn over time.	Should
Tenant (shop) management		
ADM-04	List all tenants with search and filters (plan, status, country, sign-up date, last activity, branches used).	Must
ADM-05	View a tenant profile: owner details, plan, billing history, branches, user count, usage (invoices, IMEIs, SMS), last login, support tickets and notes.	Must
ADM-06	Suspend, reactivate, extend trial, change plan, grant a free period, adjust branch limit, or archive/delete a tenant, each with a required reason recorded in the audit log.	Must
ADM-07	"Log in as" a tenant for support, only with tenant consent or a support ticket reference, shown as a banner to the admin and logged.	Should

ID	Requirement	Priority
ADM-08	Reset a tenant owner's password or 2FA after identity verification.	Must
ADM-09	Export a tenant's data on request and process data deletion requests.	Must
Plans, pricing and billing		
ADM-10	Create and edit plans: name, code, prices per term and currency, branch limit, user limit, included modules (feature flags), trial length and visibility.	Must
ADM-11	Changes to plan features apply to all tenants on that plan; price changes apply only to new subscriptions or at renewal (grandfathering).	Must
ADM-12	View all payments, invoices, refunds and failed charges; issue full or partial refunds per the refund policy.	Must
ADM-13	Record manual/offline payments (bank transfer) and activate the subscription.	Must
ADM-14	Create coupon and discount codes with value, validity, usage limit and eligible plans.	Should
ADM-15	Configure payment gateways, currencies, exchange rates and tax on subscription invoices.	Must
ADM-16	Manage SMS credit packages, top up tenant SMS balances, and view SMS usage and gateway cost.	Should
Leads, sales and affiliates		
ADM-17	List contact form leads and demo bookings with status (new, contacted, demo done, converted, lost), owner and notes.	Must
ADM-18	Manage affiliates: approve applications, set commission rates, view referrals and conversions, record payouts.	Should
Website content (CMS)		
ADM-19	Create, edit, schedule and publish blog posts with categories, featured image and SEO fields.	Should
ADM-20	Manage customer logos, testimonials, FAQs and pricing-page content without a code deployment.	Should
ADM-21	Edit SEO meta tags, redirects and sitemap settings per page.	Should
ADM-22	Edit legal pages (privacy, terms, refund) with version history.	Must
Support and communication		
ADM-23	Manage support tickets linked to tenants, with assignment, priority, status and SLA timers (priority support for Starter and above).	Should

ID	Requirement	Priority
ADM-24	Publish in-app announcements to all tenants or to a filtered group (by plan, country), with start and end dates.	Should
ADM-25	Edit system email templates (welcome, verification, trial ending, payment failed, receipt, suspension).	Must
System administration		
ADM-26	Manage platform admin users with roles: Super Admin, Support, Sales, Finance, Content Editor; 2FA mandatory for all.	Must
ADM-27	Keep a platform audit log of every admin action, searchable and exportable.	Must
ADM-28	Monitor system health: uptime, error rate, queue backlogs, SMS/email delivery failures, backup status, storage use per tenant.	Must
ADM-29	Toggle features globally or per tenant (feature flags) for staged rollouts and beta testing.	Should
ADM-30	Schedule maintenance windows and show a maintenance notice to tenants.	Should
ADM-31	Maintain global master data offered to all tenants (e.g. phone brand and model catalogue).	Could

4 Interface requirements

4.1 User interfaces

The shop application uses a left navigation menu (Dashboard, New Sale, Inventory, Purchases, Repairs, Customers, Suppliers, Finance, Staff, Branches, Reports, Settings) that collapses into a bottom or drawer menu on phones. The POS screen must be usable on a 10-inch tablet with touch targets of at least 44×44 px, and must be fully usable by keyboard and scanner alone (no mouse). Every list offers search, filters, sorting and pagination. Destructive actions ask for confirmation. The interface supports light and dark themes and English, with Sinhala and Tamil planned.

4.2 Hardware interfaces

Barcode scanners in HID keyboard mode (USB or Bluetooth); ESC/POS thermal printers and standard A4/A5 printers; cash drawers opened by printer kick pulse; device cameras for repair photos and optional camera-based barcode scanning.

4.3 Software interfaces

External system	Purpose	Method
SMS gateway (local and international)	Invoices, repair updates, reminders, OTP	HTTPS REST API, delivery-report webhook
Email service	Transactional and system email	SMTP or HTTPS API, SPF/DKIM configured
Payment gateway	Subscription payments and refunds	Hosted checkout, signed webhooks
WooCommerce	Product, stock and order sync	WooCommerce REST API with consumer key/secret, webhooks
Cloud storage	Repair photos, attachments, logos, backups	Object storage with signed URLs

4.4 Communication interfaces

All traffic uses HTTPS with TLS 1.2 or higher. Real-time dashboard and repair-status updates use WebSockets or server-sent events with polling fallback. A versioned internal REST API (JSON) serves the web client and can later be opened to partners with API keys.

5 Performance requirements

5.1 Static requirements

ID	Requirement
PER-S1	Support at least 5,000 tenants and 25,000 concurrent users on the initial infrastructure, scaling horizontally beyond that.
PER-S2	Support per tenant at least 500,000 products/IMEI records, 1,000,000 invoices and 200,000 customers without degraded search.
PER-S3	Store repair photos up to 5 MB each, compressed on upload; default storage quota per tenant set by plan.

5.2 Dynamic requirements

ID	Requirement
PER-D1	Barcode or IMEI lookup on the POS screen returns in under 300 ms (95th percentile).
PER-D2	Saving a sale invoice completes in under 1 second (95th percentile).
PER-D3	Dashboard loads in under 2 seconds on a 10 Mbps connection.
PER-D4	Standard reports for one month of data render in under 5 seconds; larger exports run in the background and notify the user when ready.
PER-D5	SMS is dispatched to the gateway within 30 seconds of the triggering event.
PER-D6	Public website pages achieve Largest Contentful Paint under 2.5 s on mobile.
PER-D7	Error rate for API requests stays below 0.1% of requests per day.

6 Design constraints

The system must be a multi-tenant cloud web application with strict tenant data isolation: every query is scoped by tenant ID, and automated tests must prove one tenant cannot read another's data. It must run in a standard browser without installation. Money must be stored as fixed-point decimals, never floating point. IMEI validation must follow the 15-digit Luhn standard. Stock quantities must be updated inside database transactions so two cashiers cannot sell the same IMEI. Plan limits (branches, modules) must be enforced on the server, not just hidden in the interface. The design must comply with Sri Lanka's Personal Data Protection Act No. 9 of 2022 and, for overseas customers, GDPR principles where applicable. Sri Lankan tax rules and printed invoice requirements apply to local tenants. Connection loss at the counter should be handled gracefully: the POS keeps the current cart and retries saving, and a full offline mode may be added later.

7 Non-functional attributes

Attribute	Requirement
Security	Encryption in transit (TLS 1.2+) and at rest (AES-256). OWASP Top 10 protections, CSRF tokens, rate limiting, parameterised queries. Annual penetration test. Secrets kept in a managed vault.
Availability	99.5% monthly uptime excluding announced maintenance, scheduled outside Sri Lankan trading hours.
Reliability and backup	Automatic daily backups kept 30 days, point-in-time recovery within 7 days; recovery point objective 24 hours or better, recovery time objective 4 hours.
Data integrity	Invoices and financial records cannot be hard-deleted, only voided with reason. Ledger balances must always equal the sum of their entries.
Scalability	Stateless application servers behind a load balancer; database read replicas and background job queues for reports, SMS and sync.
Usability	A new shop owner can make a first sale within 10 minutes of sign-up without training. Cashier tasks need no more than 3 screens.
Portability and compatibility	Works on the latest two versions of major browsers on desktop, tablet and phone.
Maintainability	Modular code by feature area, automated test coverage of at least 70% for business logic, CI/CD with staging environment.
Reusability	Shared components (tables, forms, print templates, notification service) used across modules and the admin dashboard.
Accessibility	WCAG 2.1 AA for colour contrast, keyboard navigation and form labels.
Localisation	Multi-currency display, local date and number formats, translatable interface strings.

8 Preliminary schedule and budget

This is an indicative plan for building or rebuilding the full system with a small team (1 project manager, 1 UI/UX designer, 3 full-stack developers, 1 QA engineer, part-time DevOps). Figures should be replaced with the team's own estimates.

Phase	Scope	Duration
1. Discovery and design	Confirm requirements, data model, UI design system, wireframes	3 weeks
2. Platform foundation	Tenancy, authentication, roles, sign-up, subscriptions, admin dashboard core	5 weeks
3. Core retail	Products, IMEI inventory, purchasing, suppliers, POS billing, printing, customers	7 weeks
4. Repairs and finance	Repair workflow, status page, cash drawer, banks, cheques, ledgers, expenses	5 weeks
5. Growth features	Multi-branch, transfers, staff, commissions, payroll, loyalty, credit, reports	6 weeks
6. Integrations and website	SMS, email, WooCommerce, CMS, blog, affiliates	3 weeks
7. Testing and launch	System, security and user acceptance testing; data migration; go-live	3 weeks
Total		About 32 weeks (8 months)

Cost item	Estimate (USD)
Development team (8 months, Sri Lankan market rates)	45,000 – 70,000
UI/UX design	4,000 – 7,000
Cloud hosting, storage and backups (first year)	2,500 – 6,000
Third-party services (SMS, email, payment gateway fees, monitoring)	1,500 – 4,000 plus usage
Security testing and contingency (15%)	8,000 – 13,000
Indicative total	61,000 – 100,000

9 Appendices

A Definitions and abbreviations

Term	Meaning
SRS	Software Requirements Specification
POS	Point of sale: the billing screen used at the counter
IMEI	International Mobile Equipment Identity, a 15-digit number unique to each phone
GRN	Goods Received Note: record of stock received from a supplier
Tenant	One subscribing business with its own isolated data
Super admin	Cellivo staff member with platform-wide access
MRR / ARR	Monthly / annual recurring revenue
ARPU	Average revenue per user (tenant)
RBAC	Role-based access control
2FA	Two-factor authentication
SLA	Service level agreement

B Default shop role permissions

Module	Owner	Branch manager	Cashier	Technician	Accountant
Dashboard	Full	Own branches	Own sales	Own jobs	Finance view
POS billing	Full	Full	Create	None	View
Discount over limit	Approve	Approve	Request	None	None
Returns / voids	Full	Approve	Request	None	View
Inventory	Full	Full	View	Parts view	View
Cost and margins	View	View	Hidden	Hidden	View
Repairs	Full	Full	Create	Update own	View
Finance and banking	Full	Cash drawer	Own drawer	None	Full
Staff and payroll	Full	View staff	None	None	Payroll
Settings and subscription	Full	Branch settings	None	None	None

C Main data entities

Tenant, Subscription, Plan, Payment, Branch, User, Role, Permission, Product, ProductVariant, StockItem (IMEI/serial), StockMovement, StockTransfer, Supplier, PurchaseOrder, GRN, SupplierBill, Customer, Invoice, InvoiceLine, Payment, Return, TradeIn, InstallmentPlan, LoyaltyTransaction, RepairTicket, RepairStatusHistory, RepairPart, CashSession, BankAccount, Cheque, LedgerEntry, Expense, Commission, Shift, Payroll, Notification, AuditLog, Lead, Affiliate, Coupon, BlogPost, Announcement, SupportTicket.

D Uses of this SRS

The development team builds against it; QA writes test cases from each requirement ID; support and maintenance staff use it to understand expected behaviour; the project manager bases schedule and resourcing on it; the business owner uses it to confirm what customers will get; and it can serve as the contract baseline between the developer and the client.

E Open questions for the product team

Items to confirm before sign-off: the exact free-trial length; lifetime plan terms and what "lifetime" covers; which SMS gateways and payment gateways are used; whether an offline POS mode is required; languages to support at launch; which Lite-plan reports count as "basic"; user limits per plan; and data retention after cancellation.

F References

cellivo.com public pages (home, features, pricing) as of 27 September 2026; the SRS format guide provided by the requester; IEEE 830 recommended practice for software requirements specifications.

Cellivo SRS v1.0 draft. Requirement IDs are stable and should be referenced in test cases, tickets and change requests.